

**Detecting HVAC Underperformance in Residential Buildings:
A Sensor-Based Approach to Predictive Maintenance**

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Spring 2026

ABSTRACT

Residential buildings often experience heating underperformance long before building management becomes aware of the issue. This study examines whether existing building sensor data can be used to detect poor heating performance before residents file complaints. The analysis focuses on two monitored dormitory units, A-7 and A-8, located in Building A of a Syracuse University residential complex participating in a U.S. Department of Energy research demonstration project. Using indoor temperature, occupancy, and outdoor weather data collected during the winter monitoring period, heating performance was evaluated through temperature recovery: the change in indoor temperature over a one-hour period under cold, occupied conditions. A three-condition flagging system identified heating issues when a unit was occupied, outdoor temperatures were at or below 50°F, and indoor temperature failed to increase by at least 0.9°F over a one-hour window. Sensor data were processed in Python using the pandas library and aggregated into 30-minute intervals to reduce minor short-term variations while preserving meaningful HVAC behavior. The results revealed a clear performance gap between the two units. Unit A-7 recorded a heating issue rate of 3.6%, while Unit A-8 recorded 8.6%, despite both units sharing the same building and electric resistance heating system. These findings suggest that unit-level factors, such as heat distribution, insulation quality, building orientation, or sensor placement, may influence thermal recovery behavior. More broadly, the analysis demonstrates how existing monitoring systems can support predictive maintenance in residential buildings by identifying underperformance earlier and reducing reliance on reactive maintenance practices.

Keywords: *predictive maintenance, HVAC, residential buildings, sensor data, temperature recovery, smart buildings*

1. INTRODUCTION

Residential buildings represent a significant portion of global energy use, yet their maintenance systems have failed to keep pace with advancements in building technology. Most facilities continue to rely on breakdown maintenance, addressing heating, ventilation, and air conditioning (HVAC) failures only after they occur rather than predicting and preventing issues before they affect residents. This approach results in increased energy waste, higher operating costs, and diminished resident comfort (Samatas et al., 2021). Predictive maintenance offers a data-driven option. By continuously observing real-time sensor data such as indoor temperature, humidity, and occupancy, building systems can detect performance irregularities at an early stage, prevent system failures, and reduce long-term operational expenses. In residential settings, where HVAC performance directly impacts occupants' health, comfort, and equal access to heating and indoor comfort, the impact is especially significant. Research has demonstrated that integrating real-time sensor data meaningfully improves HVAC control and system responsiveness (Tien et al., 2022). Studies on sensor-based monitoring have also shown that sensor placement can affect the accuracy of environmental data and, in turn, influence the reliability of maintenance predictions derived from that data (Samatas et al., 2021). As smart building technologies become more accessible, applying predictive maintenance approaches to residential buildings, particularly those with existing sensor systems, represents a critical and largely unexplored opportunity to improve efficiency, reduce costs, and support sustainable building management.

1.1 Background

As more buildings implement Internet of Things (IoT) sensors and machine learning models, predictive maintenance has an opportunity to improve building management, strengthening both productivity and long-term sustainability. Real-time sensor data, combined with historical maintenance records and environmental patterns, can be used to identify trends and detect issues early. Numerous studies in commercial and industrial settings demonstrate that IoT-enabled sensors, artificial intelligence, and machine learning models can successfully predict system failures, reduce energy waste, and improve maintenance schedules (Samatas et al., 2021). Machine learning techniques such as neural networks and decision trees have effectively forecasted maintenance requirements in HVAC systems, lighting controls, and building management systems (Tien et al., 2022). Predictive maintenance also aligns with green building standards, such as LEED and ASHRAE, which encourage energy-efficient operations and HVAC improvements (U.S. Green Building Council; ASHRAE). However, much of the available research focuses on large and industrial buildings with advanced, specifically designed IoT systems already in place. Residential buildings, particularly those with limited sensor availability, older construction, and varying levels of technology, remain under-studied. This gap is significant because residential settings offer ongoing challenges, including narrowed budgets, privacy concerns, and sensor limitations, that commercial-focused research does not adequately address.

Prior studies have largely focused on commercial or industrial buildings with advanced IoT systems and machine learning models designed specifically for predictive maintenance. In contrast, this study examines how predictive maintenance principles can be applied in a residential setting using sensor data that is already being collected. Rather than developing a complex machine learning model, this paper uses a simpler temperature-recovery method to demonstrate how existing residential monitoring systems can identify heating underperformance without additional infrastructure.

1.2 Problem Statement

To understand why this matters, consider a common situation: it is mid-January, outdoor temperatures have dropped to 28°F overnight, and students are returning to their dorm rooms after winter break. A functioning heating system should begin to recover to a comfortable 68–70°F within an hour of occupancy. When it fails to do so, students return to cold rooms, and under the current maintenance approach, building staff only become aware of issues after a complaint is filed. Even then, resolving the issue can take an additional one to two hours, leaving residents uncomfortable for a prolonged period before the problem is addressed. Predictive maintenance offers a

fundamentally different approach. Rather than waiting for a complaint, a predictive system continuously monitors real-time sensor data – temperature recovery rates, outdoor conditions, and occupancy patterns – and automatically flags when a unit's heating response falls below expected performance. Building managers receive an alert, maintenance can investigate early, and the issue can be resolved before the student even comes back.

1.3 Aims of the Study

This study examines how existing sensor data can support predictive maintenance in university residential buildings by identifying patterns of heating underperformance before residents file complaints. The analysis focuses on two residential dormitory units, Unit A-7 and Unit A-8, located in Building A of a university residential complex at Syracuse University, as part of an ongoing research demonstration project sponsored by the U.S. Department of Energy. Since both units share the same building and central heating system, they provide a controlled comparison, in which differences in heating performance are more likely to reflect unit-level factors such as building position, insulation quality, or heat distribution rather than a system-wide issue. Unit A-7 is monitored by a Front Door sensor, whereas Unit A-8 is monitored by a Kitchen sensor; these differences in sensor placement are an important consideration when interpreting results. The goal of this study is to measure and compare the heating performance of the two units using temperature recovery as the primary metric: the increase in indoor temperature over a 1-hour period under cold, occupied conditions. By identifying patterns of weak or insufficient heating recovery, this study demonstrates how existing sensor data can support predictive maintenance by detecting underperformance early, before it becomes a larger issue or a resident complaint.

This study addresses the following research questions:

- To what extent can temperature recovery data serve as an indicator of HVAC underperformance in residential units?
- What patterns of heating inefficiency are observable between units sharing the same central heating system?
- In what ways can existing building sensor data be utilized to support a predictive maintenance framework without additional infrastructure?

2. BUILDING AND SENSOR OVERVIEW

2.1 Study Site

This study is part of a residential building performance research project at Syracuse University, located on South Campus in Syracuse, New York (Mirzabeigi et al., 2025). It is part of a larger "living laboratory" effort funded by the U.S. Department of Energy (DOE) Advanced Building Construction (ABC) initiative (Syracuse University Magazine, 2025). The focus is on evaluating energy efficiency, indoor environmental quality (IEQ), and overall building performance in cold-climate residential settings through data-driven analysis (Syracuse University Magazine, 2025). The research is conducted in occupied student housing, allowing building systems to be observed under real-world conditions (Mirzabeigi et al., 2025). This approach enables analysis of how building performance interacts with occupant behavior, capturing detailed patterns of energy use, indoor conditions, and system response in an active residential setting (Mirzabeigi et al., 2025). This section focuses on Building A, one of two nearly identical two-story apartment buildings originally constructed in the early 1970s. Buildings of this type are common in the northeastern United States and are often associated with heat loss and air infiltration. Building A consists of eight residential units, each with existing conditions that pose several technical challenges for maintaining thermal comfort and efficiency. The units utilize electric resistance heating, an energy-intensive method that is highly sensitive to external climate conditions and often results in high utility costs during Syracuse winters. (Mirzabeigi et al., 2025). The buildings have no mechanical cooling system installed, leaving the units dependent on natural ventilation or window units during warmer periods. The buildings also lack reliable mechanical ventilation; fresh air

is exchanged primarily through natural infiltration through the building enclosure and by residents manually opening windows. A key operational constraint is that occupants have no direct control over the heating system. Syracuse University Facilities maintains a set temperature of 73°F, with the system typically deactivated from mid-April to mid-September (Mirzabeigi et al., 2025). Together, these conditions, including inefficient heating and limited insulation, establish the context for evaluating pre-retrofit performance. This study focuses on the winter heating performance of Units A-7 and A-8 in Building A during the pre-retrofit period.

2.2 Sensor System and Data Collection Framework

Building A is provided with a real-time monitoring system designed to capture multiple aspects of building performance simultaneously (Mirzabeigi et al., 2024). The sensor system includes Milesight AM319 monitoring sensors that measure indoor air temperature, relative humidity, carbon dioxide (CO₂), total volatile organic compounds (TVOCs), and particulate matter (PM2.5 and PM10). Occupancy is tracked using Netvox RB11E passive infrared (PIR) motion sensors installed at unit entrances, while Netvox R313A door and window contact sensors record when openings happen, providing insight into occupant behavior and its effect on indoor conditions and heat loss. Energy consumption at both the building and unit levels is measured using eGauge power meters at one-minute intervals. Building enclosure performance is monitored using heat flux sensors, surface temperature sensors, and local air temperature and humidity sensors installed at 10 locations across the building's interior and exterior walls (Mirzabeigi et al., 2024). Outdoor weather data, including temperature, humidity, solar radiation, and wind, are collected from an on-campus weather station and integrated into the central database (Mirzabeigi et al., 2025). All sensors use the LoRaWAN wireless communication protocol, a technology selected for its long-range communication capability and low power usage, given that campus Wi-Fi access is restricted for security reasons (Mirzabeigi et al., 2024). Sensor data is routed through a LoRa gateway and logged to a central SQL database via two separate Python processes: one for energy meters and one for environmental and occupancy sensors. This system ensures that failures in one data source do not disrupt others, maintaining continuous monitoring (Mirzabeigi et al., 2025). Indoor environmental and occupancy data are collected at five-minute intervals, providing high-resolution time-series data across all monitored units.

2.3 Description of Selected Research Units

Among the eight units in Building A, Units A-7 and A-8 are designated as the primary research units. They are equipped with a more extensive sensor system than the typical apartments (Mirzabeigi et al., 2025). All units are monitored for indoor air quality and occupancy at the living room level, while the research units have additional sensors in the kitchen and both bedrooms. These include door and window contact sensors that record detailed interactions by occupants with the building. This expanded sensor setup makes A-7 and A-8 the most thoroughly monitored units in the building and enables more detailed analysis. Unit A-7 is monitored by a Front Door (FD) sensor positioned near the unit entrance, while Unit A-8 is monitored by a Kitchen (KT) sensor located adjacent to the thermostat (Mirzabeigi et al., 2025). These differences in sensor placement are important when interpreting temperature readings. The kitchen sensor may be influenced by internal heat sources, such as appliances. The front door sensor is more exposed to cold air infiltration from the hallway or exterior. Since both units share the same building and central heating system, differences in measured heating performance are more likely to reflect unit-level factors, such as physical position within the building, insulation quality, or heat distribution, rather than a system-wide issue. This makes them appropriate for comparative analysis in this study.

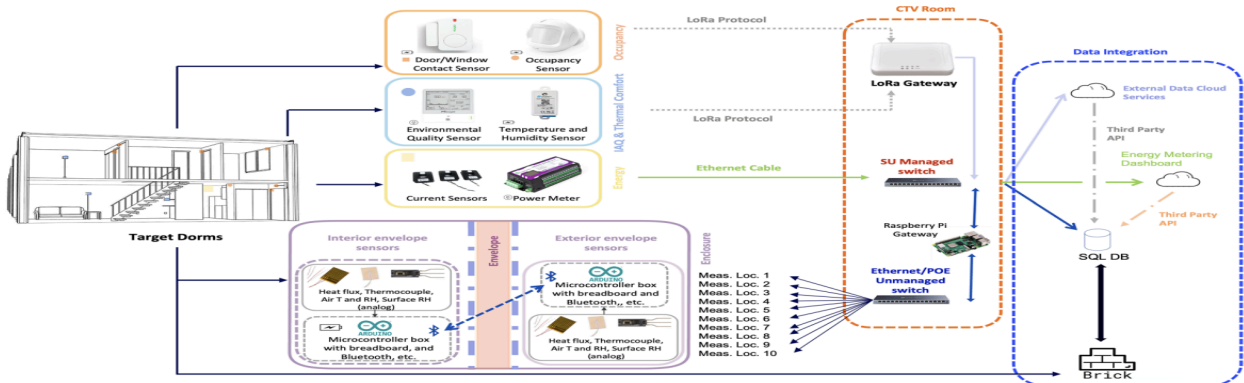


Figure 1 illustrates the full system architecture for real-time performance monitoring, including sensor types, communication pathways, and data integration infrastructure (Mirzabeigi et al., 2024)

2.4 Site Observations and Field Context

A site visit to Building A provided additional context regarding the monitoring systems and HVAC infrastructure described in the project documentation. Motion sensors, door and window contact sensors, environmental monitoring devices, thermostats, and HVAC-related ductwork were directly observed throughout the building during the visit. The observations confirmed the implementation of a real-time monitoring network designed to track occupancy, indoor environmental conditions, and building system behavior. Smart thermostat systems and automated HVAC controls observed on-site were consistent with the building's broader energy management strategy described in prior research (Mirzabeigi et al., 2025). The visit also provided a better understanding of how sensor placement, occupant interaction, and ventilation systems may influence indoor thermal conditions and heating performance within the residential units.

3. METHODOLOGY

3.1 Data Preparation

The analysis uses two data sources from the winter monitoring period. Indoor sensor data from Units A-7 and A-8 capture temperature and occupancy via motion detection. Outdoor weather data is from the on-campus station. The dataset is from Mirzabeigi et al. (2025). Preprocessing and analysis were completed in Python with the pandas library. Prior to analysis, each dataset required cleaning and formatting. The indoor sensor files contained timestamps recorded in Eastern Standard Time with a trailing " EST" suffix, which was removed before parsing. Timestamps were then converted into a standardized datetime format to ensure consistent parsing across both unit files. Any rows where timestamp parsing failed were dropped from the dataset, and all records were sorted chronologically before further processing. Temperature values from the indoor sensors were originally recorded in degrees Celsius. These were converted to degrees Fahrenheit using the standard conversion formula ($^{\circ}\text{F} = ^{\circ}\text{C} \times 9/5 + 32$) to align with U.S. building management practices and to allow direct comparison with outdoor weather data. For the outdoor weather dataset, a mean-value check was applied to the thermometer column to determine whether values were already in Fahrenheit or required conversion to Fahrenheit. If the column mean fell below 45, the data was treated as Celsius and converted accordingly. To merge the indoor and outdoor datasets, a nearest-timestamp join was performed with pandas `merge_asof`. This pairs each indoor reading with the closest outdoor weather record within a 30-minute window. The approach accounts for different recording intervals and ensures each indoor observation is matched to a representative outdoor temperature without large temporal gaps.

3.2 Defining Heating Performance

Heating performance in this study is defined as temperature recovery, specifically the increase in indoor temperature over time while the unit is occupied under cold outdoor conditions. Rather than evaluating whether a unit reaches a specific target temperature, this analysis focuses on the rate of thermal recovery as the primary indicator of system responsiveness. A well-functioning heating system should produce a measurable, consistent rise in indoor temperature within a reasonable timeframe after occupancy or a drop in outdoor temperature. The 1.8°F per hour baseline was chosen based on the typical temperature rise observed during stable heating periods across the full winter dataset and was used as a reference point for expected heating performance.

This definition is grounded in the target comfort range of 68–70°F commonly referenced in residential building management. As outdoor temperatures decrease, maintaining this indoor range becomes increasingly demanding because the heating system must sustain a larger indoor-outdoor temperature differential. This analysis focuses on

periods when outdoor temperatures are at or below 50°F, as these conditions yield more consistent heating demand and enable a more reliable evaluation of differences in unit-level heating performance.

3.3 Temperature Aggregation and Recovery Metric

The raw sensor data for both units is recorded approximately every five minutes. While this resolution captures short-term fluctuations in indoor conditions, it introduces noise that is not representative of actual HVAC system behavior. Heating systems do not respond instantaneously; they operate in cycles, requiring time to push conditioned air through the space and produce a measurable change in room temperature. Analyzing temperature changes at five-minute intervals can capture brief changes unrelated to heating performance, such as a door opening or a sensor reading anomaly. For this reason, data was aggregated into 30-minute intervals, which smooths out minor changes while maintaining enough temporal resolution to track significant system behavior throughout the day. The core metric used in this analysis is the one-hour temperature change, referred to throughout as the "delta." This measures how much the indoor temperature rises over a 60-minute window during cold, occupied conditions. A one-hour window was selected because it balances two competing concerns: windows that are too short capture noise rather than meaningful trends, while windows that are too long, such as two to three hours, introduce too many additional variables, including shifts in outdoor temperature and changes in occupancy. One hour is a realistic and widely applicable benchmark for thermal recovery in residential heating. Since sensor readings are available every 30 minutes after aggregation, a nearest-timestamp merge was applied with a 30-minute tolerance window to pair each observation with its corresponding reading approximately one hour later, producing a reliable delta value for each time step.

3.4 Heating Issue Classification

A heating issue is flagged when three conditions are met simultaneously: the unit is occupied, as determined by motion-sensor data; the outdoor temperature is at or below 50°F; and the indoor temperature does not increase by at least 0.9°F over the one-hour window. All three conditions must be present simultaneously for a flag to be recorded. An unoccupied unit or mild outdoor temperatures would not trigger the classification, regardless of the temperature delta. The 0.9°F threshold represents the minimum expected thermal recovery for a functioning system under cold conditions. As established in Section 3.2, the observed baseline for effective heating in this dataset is approximately 1.8°F per hour. A delta below 0.9°F, roughly half the expected rate, indicates a heating response that is significantly weaker than normal. A one-hour delta between 0.9°F and 1.8°F may suggest slower-than-expected performance and is worth monitoring, while anything below 0.9°F is classified as a heating issue for the purposes of this analysis. A heating issue flag should not be interpreted as a direct indication of system failure. Rather, it indicates that thermal recovery during that hour fell below the expected performance threshold. Isolated flags may reflect temporary conditions such as a window being left open or an unusual occupancy pattern. However, when flags cluster together, particularly during the coldest periods of the winter season, they form a reliable early warning signal of sustained underperformance, a pattern that a predictive maintenance system could detect and address before a resident complaint is ever filed.

3.5 Analysis Process

The analysis was conducted in Python using the pandas library, following a consistent analytical workflow for both Unit A-7 and Unit A-8. First, the raw sensor files were loaded, the "EST" suffix was removed from the timestamps, and the timestamp column was converted into a consistent datetime format using a specified format string. Any rows that failed parsing were dropped, and all records were sorted chronologically. Temperature values were converted from Celsius to Fahrenheit, and a binary occupancy column was derived from the motion sensor's 'detected' or 'not-detected' status. Following data preparation, a one-hour temperature delta was computed by matching each observation with the closest corresponding reading approximately one hour later using a nearest-timestamp merge with a 30-minute tolerance window. The resulting delta column captures the change in indoor temperature over that

hour. The outdoor weather dataset was then merged onto the indoor records using the same nearest-timestamp approach, aligning each indoor observation with the closest available outdoor temperature reading. Once all variables were prepared, three binary flag columns were generated to classify heating conditions. The primary classification logic used in this study is shown below:

```
occ['heating_issue_cold'] = (
    occ['cold'] &
    (occ['occupied'] == 1) &
    occ['has_delta'] &
    (occ['delta_1hr'] < 0.9)
)
```

This single line captures all three conditions: cold outdoor temperature, confirmed occupancy, and insufficient thermal recovery, producing a Boolean flag for each observation period. Flagged rows were then exported for further analysis and visualization, and summary statistics were calculated to determine the overall heating issue rate for each unit over the full winter monitoring period.

3.6 Challenges and Resolutions

Several challenges arose during the analysis, requiring adjustments to the methodology—the most significant involved timestamp alignment. The raw indoor sensor files contained a trailing "EST" timezone suffix that interfered with standard datetime parsing. This issue was resolved by removing the suffix before converting timestamps into a consistent datetime format. Another challenge involved merging datasets collected at different recording intervals. To address this, a nearest-timestamp merge with a 30-minute tolerance window was used to align indoor sensor, occupancy, and outdoor weather records while keeping timestamps reasonably synchronized across datasets. Selecting cutoff values for heating issue classification also required refinement. Early testing with different delta values and outdoor temperature conditions produced results that either flagged too many hours or failed to capture meaningful heating patterns. The final thresholds of 0.9°F and 50°F were selected based on the overall distribution of temperature changes observed in the dataset and through discussion with the research team. To support the quantitative analysis, a short survey was conducted with facilities staff involved in building operations and maintenance. Responses provided additional context on how sensor data is currently accessed and highlighted the lack of an automated alert system for identifying HVAC-related issues. These findings informed the recommendations and predictive maintenance discussion presented later in the paper. Throughout the analysis, discussions with the faculty research mentor helped refine the methodology, the selection of cutoff values, and the interpretation of the results.

4. RESULTS

The analysis examined all occupied hours during the winter monitoring period in which outdoor temperatures were at or below 50°F, covering the full cold-weather season for both units. The results reveal a clear and consistent performance gap between Unit A-7 and Unit A-8.

Table 1. Heating issue rates — Units A-7 and A-8, winter season (outdoor temp ≤ 50°F, occupied hours only)

Unit	Sensor location	Cold occupied hours	Issues flagged	Issue rate
A-7	Front Door (FD) sensor	7,671	275	3.6%
A-8	Kitchen (KT) sensor	8,560	732	8.6%

A heating issue was defined as an occupied hour with outdoor temperature ≤ 50°F in which indoor temperature increased by less than 0.9°F over a one-hour period.

Unit A-7, monitored by the Front Door sensor, recorded a heating issue rate of 3.6%, meaning that out of 7,671 cold-occupied hours analyzed, 275 hours showed weak thermal recovery. Unit A-8, monitored by the Kitchen sensor, recorded a heating issue rate of 8.6%, with 732 flagged hours out of 8,560. Although both units share the same building and electric resistance heating system, Unit A-8 had more than twice as many heating issues as Unit A-7, suggesting that conditions within each unit may be influencing heating performance differently. Figures 2 and 3 further illustrate differences in heating recovery behavior between the two units during the February 1–8 monitoring period. Unit A-7 shows relatively few isolated flagged heating periods throughout the monitoring window, whereas Unit A-8 exhibits more frequent, recurring flagged heating periods over the same timeframe. The visual difference between the two figures supports the broader winter-season findings in Table 1, suggesting that weaker thermal recovery in Unit A-8 was not limited to isolated occurrences but occurred more regularly throughout the monitoring period. Across the full winter monitoring season, the 8.6% heating issue rate observed in Unit A-8 corresponds to hundreds of occupied hours with weak thermal recovery during cold weather. In contrast, Unit A-7 showed substantially fewer flagged periods across a similar number of occupied hours. These findings suggest that measurable differences in heating performance can be identified using existing building sensor data and may support earlier detection of residential HVAC underperformance.

Figure 2. Unit A-7 (Feb) — Indoor vs Outdoor Temperature with Slow Heating Flags

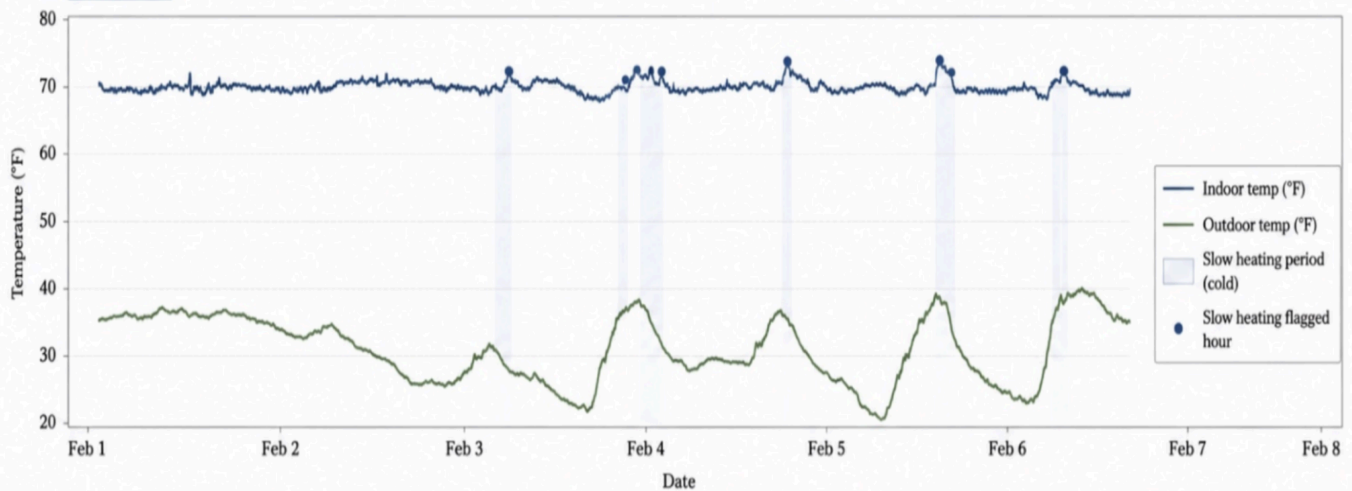
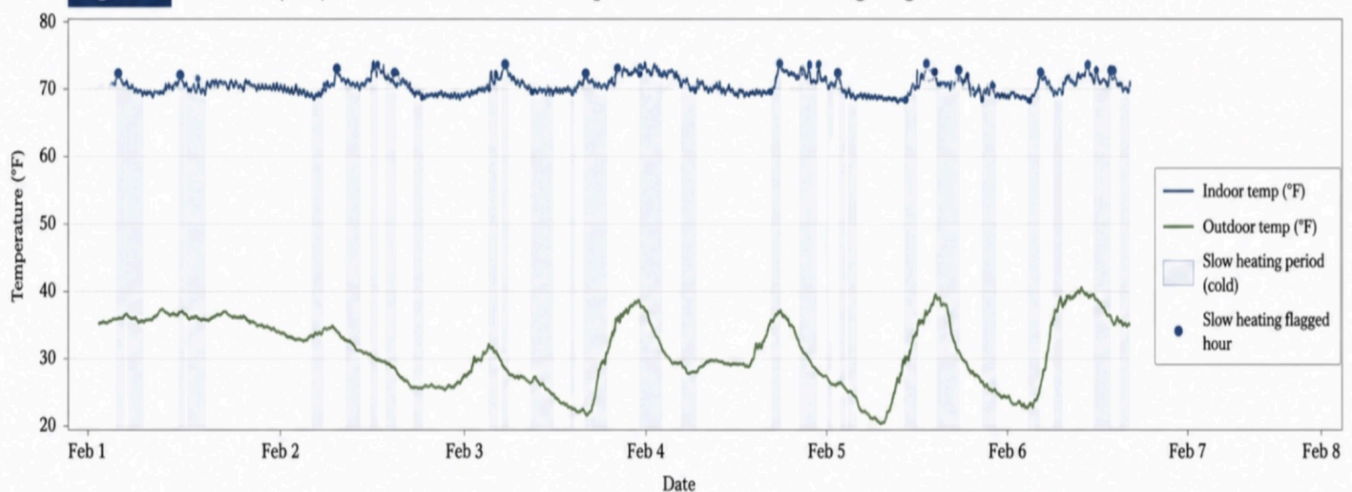


Figure 3. Unit A-8 (Feb) — Indoor vs Outdoor Temperature with Slow Heating Flags



Note. Shaded regions indicate hours when outdoor temperature $\leq 50^{\circ}\text{F}$ and indoor temperature increased by less than 0.9°F over a one-hour period.

5. DISCUSSION

5.1 Explanations for the Performance Gap

The results indicate a heating issue rate of 8.6% in Unit A-8, compared to 3.6% in Unit A-7, even though both units are located in the same building and use a shared electric resistance heating system. This disparity likely results from several contributing factors rather than a single cause. Building position appears to be one of the most likely explanations. As shown in Figure 1, Building A is a two-story attached residential structure located on the southeastern edge of South Campus. Units positioned at the ends or corners of a building are exposed to more exterior wall surface area than interior units, increasing heat loss and placing greater demand on the heating system to maintain stable indoor temperatures. If Unit A-8 is located at an end or corner relative to A-7, this may contribute to the observed performance gap.

Differences in localized heat distribution may also help explain variations in thermal recovery behavior between the two units. Because the building relies on electric resistance baseboard heating, heat delivery depends heavily on the location of the heating elements, the thermostat's responsiveness, and the unit's ability to retain heat. If Unit A-8 experiences greater heat loss through exterior walls, windows, or doors, indoor temperatures may recover more slowly in cold weather, even when the heating system remains active. Differences in insulation quality and air sealing around windows and exterior doors may further contribute to slower thermal recovery and increased heating demand. Based on the consistency of the observed heating patterns across the monitoring period, building position, and localized heat loss, these appear to be the most plausible physical explanations for the performance gap between the two units.

Beyond these physical factors, sensor placement introduces an additional methodological limitation. Unit A-7 was monitored with a Front Door sensor near the unit entrance, while Unit A-8 was monitored with a Kitchen sensor closer to the thermostat and internal heat-generating appliances. Because these sensors were positioned in different thermal environments, they may have captured location-specific temperature behavior differently. Occupancy behavior, such as more frequent door openings, may also contribute to increased heat loss, though further occupancy analysis would be needed to evaluate this connection fully. Sensor placement remains an important limitation of this study. The Front Door sensor in Unit A-7 was positioned near a potential source of cold-air infiltration, while the Kitchen sensor in Unit A-8 was located in a space that may be influenced by appliance-related heat gains and proximity to the thermostat. These differing thermal conditions may partially or meaningfully explain the observed performance gap between the two units. Future work using standardized sensor placement across all monitored units would help more reliably isolate HVAC performance effects.

5.2 Predictive Maintenance Application

The heating performance patterns identified in this study have direct implications for predictive maintenance. Facilities staff involved in building operations noted that no automated alert system currently exists for identifying units with recurring heating underperformance. As a result, maintenance responses still depend heavily on resident complaints, meaning issues are typically addressed only after residents report discomfort. This reinforces the value of an automated flagging system that can identify repeated underperformance earlier using existing sensor data. When a unit's thermal recovery consistently falls below the expected threshold during cold, occupied conditions, it may indicate that part of the heating system or building enclosure is underperforming. Because these patterns can be automatically detected from sensor data, building management can identify issues before residents submit maintenance complaints.

In practice, a predictive maintenance system based on this analysis would continuously monitor thermal recovery rates, outdoor conditions, and occupancy patterns. When flagged hours occur more frequently, the system could notify building management so the unit can be inspected further. Baseboard heating elements and thermostat

controls should be prioritized for inspection when repeated slow-heating patterns are detected. Blocked airflow around heating units, older heating elements, thermostat issues, or heat loss near windows and exterior doors may contribute to slower temperature recovery even when the heating system is still operating normally. By identifying recurring periods of weak thermal recovery in sensor data, building management can target inspections earlier and address issues before they lead to larger system failures. This approach shifts building management from reactive to proactive. Rather than waiting for residents to report uncomfortable indoor conditions, underperformance patterns can be identified earlier, before additional energy loss occurs and before heating performance declines further.

5.3 Data Requirements for Predictive Maintenance

Implementing a predictive maintenance system, as discussed in this study, requires consistent sensor data in three main categories: indoor temperature, occupancy status, and outdoor weather conditions. Indoor temperature is the most critical variable. It is used to calculate thermal recovery rates and to identify periods of underperformance. This study demonstrates that five-minute sensor readings aggregated into 30-minute intervals provide sufficient resolution to capture meaningful HVAC behavior while reducing **minor short-term variations**. Similarly, the one-hour analysis window balances realistic system response with the need to limit the influence of changing outdoor conditions or occupancy behavior over longer periods. A reliable predictive maintenance system also requires enough historical data to establish normal heating behavior for each unit. Without a baseline, it becomes difficult to determine what qualifies as underperformance. In this study, the observed distribution of thermal recovery values across the winter monitoring period was used to establish the 1.8°F/hour baseline and the 0.9°F flagging threshold. For buildings without prior monitoring history, at least one full heating season of data would likely be needed before reliable thresholds could be established.

5.4 Recommendations

Based on the findings of this study, several recommendations are proposed for building management and future research. The findings suggest that Unit A-8 may benefit from further physical inspection. With a heating issue rate of 8.6%, the unit consistently exhibited weaker thermal recovery than Unit A-7 under similar outdoor and occupancy conditions. Inspection efforts should prioritize baseboard heating components, thermostat performance, window and door air sealing, and insulation quality, particularly if A-8 occupies an end or corner position within the building. Both units should continue to be monitored during future winter seasons. While a single heating season provides a useful baseline, evaluating whether the performance gap between A-7 and A-8 persists over time would strengthen the interpretation of the results. Monitoring could also be expanded to additional units within Building A to determine whether similar heating patterns exist elsewhere in the building. Finally, the automated flagging methodology developed in this study could be integrated into the building's existing monitoring platform to support earlier identification of heating underperformance without requiring continuous manual analysis.

Future implementation efforts should focus on integrating automated flagging systems into existing building monitoring platforms. Rather than relying solely on resident complaints, building management systems could continuously evaluate thermal recovery behavior and generate alerts when repeated patterns of underperformance are detected. Expanding this approach to additional residential buildings may support earlier maintenance interventions, improved thermal comfort, and more efficient building operations over time.

6. CONCLUSION

This study explored whether existing building sensor data could be used to identify heating underperformance in residential buildings before residents file complaints. Using a temperature recovery analysis applied to monitoring data from two residential dormitory units at Syracuse University, the study identified a consistent performance gap between Units A-7 and A-8, despite both units sharing the same building and electric resistance heating system. The

results demonstrated that temperature recovery can serve as a practical indicator of heating performance under cold, occupied conditions. More importantly, this study demonstrates a replicable approach to identifying residential heating underperformance using sensor systems already installed in many monitored buildings. Rather than relying on complex machine learning models, the methodology developed here uses temperature recovery patterns to identify recurring underperformance using existing environmental and occupancy data. Unit A-8 exhibited more than double the heating-issue rate of Unit A-7, suggesting that unit-level factors, such as building orientation, heat-loss insulation quality, or sensor placement, may influence thermal recovery behavior. Importantly, these patterns were detected using existing monitoring systems without requiring additional equipment or physical inspection methods. More broadly, the findings demonstrate how real-time sensor data can support predictive maintenance in residential buildings. By automatically identifying recurring periods of weak thermal recovery, building management can detect underperformance earlier, prioritize inspections more effectively, and reduce reliance on reactive maintenance. As residential buildings continue to integrate real-time monitoring systems, approaches such as those developed in this study may support more proactive, data-informed building management practices.

6.1 Future Work

This study opens several avenues for future research on building performance, predictive maintenance, and residential energy efficiency. One possible next step is to extend the analysis to the summer cooling season to determine whether similar performance differences occur under cooling conditions and whether the same methodology can detect underperformance in a different thermal setting. Expanding the analysis to all eight units in Building A and, eventually, Building B would also provide a broader view of building-wide heating performance and a larger dataset for establishing more reliable baselines. Longer-term monitoring over multiple years would help distinguish between year-to-year variation in outdoor conditions and actual changes in system performance. This would be particularly useful when evaluating how future retrofits affect heating efficiency and thermal recovery behavior. The flagging methodology developed in this study also provides a foundation for future machine learning applications. With sufficient historical data across multiple units and heating seasons, supervised learning models could potentially be trained to identify units likely to experience heating underperformance before issues become persistent. Establishing reliable unit-level baselines would be an important prerequisite for this type of predictive modeling. Future implementation efforts could also focus on integrating automated flagging systems into existing building monitoring platforms. Real-time dashboards displaying heating-issue rates, flagged hours, and outdoor temperatures may help make these analyses more accessible to building managers and facilities staff without requiring advanced technical expertise. Finally, because the dataset used in this study is publicly available from Mirzabeigi et al. (2025), other researchers may be able to apply and extend this methodology to different residential buildings, climate conditions, and occupancy contexts. Expanding these approaches across larger datasets may help support more scalable predictive maintenance strategies in residential building management over time.

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